for about a week, during which sugars and other simple carbohydrates are rapidly metabolized. This is an exothermic process and may cause an increase in temperature by 40°C. Example: *Bacillus subtilis*

2. **Thermophilic stage** - Thermophilic is the second stage, lasting for about two weeks, during which the temperature may rise to about 50 to 75°C. Such a drastic increase in temperature is accompanied by the decomposition of cellulose and other resistant materials. It is important that the material be thoroughly mixed and kept aerated during this stage. Example: *Bacillus stearothermophilus*
3. **Curing stage** - The temperature decreases during this final stage and the material being composted is recolonized by mesophillic organisms, which often produce plant-growth stimulating compounds.

The humification of organic material is characterized by an increase in concentration of humic acids approximately from 4 to 12 percent, and decreases during the composting process.



What should you compost?

When selecting materials for your compost pile avoid the following:

- Wastes that attract pests
- Diseased / insect ridden plants
- Non-biodegradable things



Compost bed types

1. Pit method
2. Heap method

Pit method

The compost pits dug in soil with dimension of 3.5m × 2.5m × 1.5m (L×B×H). The pits are filled layer by layer using green plants and animal excreta. The layering is repeated until the pit is filled. Finally a layer of mud is plastered on the top of the pit (Figure 9.12).



Figure 9.12: Pit method

Heap method

In regions with heavy rainfall, the compost may be prepared in heaps above the ground level and protected by a shed. The pile is made with dimension of 2m × 2m × 1.5m (L×B×H) (Figure 9.13).



Figure 9.13: Heap method

Methods of compost preparation

1. Indore method
2. Bangalore method

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